MIL-S-23378C
24 October 1978
SUPERSEDING
MIL-S-23378B
8 March 1974

MILITARY SPECIFICATION

SCREW MACHINES, AUTOMATIC, SINGLE SPINDLE 1/2 INCH THROUGH 2-3/8 INCH SPINDLE CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single spindle, turret type screw machines capable of automatically producing turned and threaded parts from barstock.

1.2 Classification. The automatic screw machines covered by this specification are of the following classes and sizes. The class and size to be supplied shall be as specified by the procuring activity (see 6.2.1).

Class 1 - Main drive motor not less than 3 HP
Maximum turning length not less than 1-1/2 inches

Size - 1/2 - Maximum bar diameter 1/2 inch
3/4 - Maximum bar diameter 3/4 inch

Class 2 - Main drive motor not less than 7-1/2 HP
Maximum turning length not less than 3-1/2 inches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3416

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

Size - 3/4 - Maximum bar diameter 3/4 inch
Size - 1-1/4 - Maximum bar diameter 1-1/4 inch
Size - 1-5/8 - Maximum bar diameter 1-5/8 inch
Size - 2 - Maximum bar diameter 2-3/8 inch
Size - 2-3/8 - Maximum bar diameter 2-3/8 inch

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058	Machinery, Metal and Woodworking, Preparation for Delivery of

STANDARDS

FEDERAL

FED-STD-595	Colors
FED-STD-28	Screw Thread Standards for Federal Services

MILITARY

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments,	Occupational Safety and Health Administration Standards

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

* NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 79 - Electrical Standard for Metalworking Machine Tools

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

* AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z55.1 - Gray Finishes for Industrial Apparatus and Equipment

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the contractor shall furnish a sample unit(s) for first article inspection and approval (see 4.2 and 6.3).

3.2 Design. The machine shall be the manufacturer's current model designed to perform all machine functions automatically, including turning, forming drilling, threading, cut-off and bar feed. Each tool slide shall be independently operated so that machining operations can be performed simultaneously and in any sequence. Provisions shall be made for readily mounting available attachments and accessories.

3.2.1 Safety and health requirements. The machine shall have covers or other safety devices for all parts that present safety hazards. The devices

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

shall include guards for belts, pulleys, chains, gears, and sprockets. All guards shall be removable to provide easy access to the guarded parts, and shall not interfere with operation of the machine. Safety devices shall be provided to protect the machine drive mechanisms from damage by overload, to prevent drive shaft operation when the feed gear cover is open, and to prevent power operation when the handwheel is engaged. The machine shall comply with the standards promulgated under OSHA which are applicable to the machine. Additional requirements for safety and health shall be as specified (see 6.2.1 and 6.4).

3.2.2 Lubrication. An automatic lubricating system shall provide positive lubrication to all enclosed mechanisms and slide surfaces requiring continuous lubrication. Recirculating lubricating systems shall have a replaceable or cleanable filter. Each oil reservoir shall be provided with an oil level indicator. Parts lubricated manually shall have easily accessible oil grease fittings.

3.2.3 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances which will permit installation of replacement parts without modification or parts or machine.

3.3 Construction. The machine shall be new and constructed of parts which are free of defects and repair, and of the necessary quality to conform to the standards for construction, safety and accuracy specified herein. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, cooper, brass, aluminum, fiber products, plastics and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse or sludge.

* 3.3.2 Castings and forgings. All castings and forgings shall be free of scale and mismatching. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Painting. Unless otherwise specified, the machine shall be painted in accordance with the manufacturer's commercial practice. When specified (see 6.2.1), the final color shall be Standard Machine Tool Light Gray similar to No. 16187 Gray Gloss in accordance with FED-STD-595 or ANSI Z55.1, Chip No. 70, light gray. Pinch points and other physical hazards on the machine shall be color coded in accordance with paragraph 1910.144 of OSHA.

* 3.3.5 Threads. All threads shall conform to FED-STD-H28.

3.3.6 Dials. All dials shall be graduated in the inch system. Graduations indicating stock removal or tool movement shall be in increments not greater than 0.001 inch. Dial diameters shall be such that graduations may be easily read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

* 3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background. Danger, caution, and safety instructions shall conform to paragraph 1910.145 of OSHA.

3.4 Components.

3.4.1 Bed and base. The machine bed shall be constructed to protect the slide mechanisms from chips, dirt, and coolant. The chip compartment shall have a perforated divider to separate chips from coolant being returned to the reservoir. Slide ways for supporting and guiding the indexing turret slide and cross slides shall have replaceable steel ways having a surface hardness not less than Rockwell C-60. A removable tray shall be mounted in the bed for catching completed work parts. The base of the machine shall be equipped with means for securing and leveling the machine on a foundation.

3.4.2 Spindle and spindle drive. The spindle shall be designed for advancing bar stock through the spindle. The spindle nose shall be designed for readily changing collets. The spibdle drive mechanism shall rotate the spindle in both directions by a multiple chain and gear drive. Means for

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

adjusting chain tension shall be provided. Speed changes shall be made automatically during the machining cycle. Spindle speeds and speed ratios, shall be obtained by changing gears to provide the required speeds within the range of the machine. The spindle shall be mounted in preloaded ball or roller bearings. All rotating shafts in the drive mechanism shall be mounted in ball or roller bearings. A metal plate shall be permanently mounted convenient to the gear box showing the gear combinations for obtaining the available speeds.

3.4.3 Stock feeding mechanism. The stock feeding mechanism shall automatically advance bar stock through the spindle to a swing stop. The stock stop shall be adjustable for part length and timing to operate with the stock feed mechanism. The bar shall be advanced any length in one movement within the capacity of the machine. The feed mechanism shall have means for gaging the stock feed stroke and for adjusting the grip on the bar diameter. The feed mechanism drive sleeve speed shall be independent of the main drive shaft speed and shall be provided with speed change gears. An electrically operated stopping device shall stop the machining cycle and give a warning signal when a bar of stock is expended.

3.4.4 Indexing turret. The standard indexing tool turret shall have six index positions with provision for double indexing to alternate positions. The tool turret shall be mounted on the turret slide in ball or roller bearings. The turret shall be designed for mounting interchangeable round shank tool holders. The turret slide shall provide feed movement by cam action. Provisions shall be made to compensate for wear on the turret slide ways. The turret slide stroke shall have a longitudinal screw adjustment to suit various cam lobe heights and tooling. The turret slide shall be equipped with a rapid pull-out-and-return mechanism for chip clearance during deep drilling operations. Each Class 2 machine shall have a positive stop to limit forward motion of the slide.

3.4.5 Cross slides and upper rear slide. Cross slides shall be mounted on the horizontal bed ways at the front and rear of the spindle and an upper rear slide shall be mounted above the spindle, each providing tool feeds at a right angle to the spindle axis. Each slide shall be cam operated. An adjustment screw with a dial graduated in 0.001 inch increments shall be provided for transverse adjustment on each cross slide. Cross slides shall have standard size T-slots for mounting interchangeable tool holders. The vertical slide shall be for clamping various size cut-off blades. Each cross slide and the vertical slide shall operate independently and in combinations. Each Class 2 machine shall have a micrometer adjustment to set the cut-off slide.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

3.4.6 Operating mechanisms. Cam holders for mounting the slide operating cams shall be graduated and shall be adjustable for angular position. Mechanical devices for automatically actuating machine functions shall be adjustable for positioning, or shall be graduated drum type carriers. Automatic functions shall be performed in any required sequence of operation including: turret index, stock feed, spindle speed change and spindle reverse, work deflector, and auxiliary functions as provided. Manually operated electrical controls shall include spindle start and stop, drive start and stop, and jog functions. The main drive shaft shall operate at a constant speed as established by change gears. A handwheel shall be provided to cycle the machine during set-up. A safety interlock shall prevent the driveshaft from operating unless the handwheel is disengaged. Each slide shall be provided with means for hand operation during set-up. The drive shaft shall be operated by a separate electric motor.

3.4.7 Change gears. Gears shall be changeable without disassembling any components of the machine. The gears shall be locked on the gear shafts with means to prevent gears working loose during operation. A compartment shall be provided in the machine base for storing gears not in use .

3.5 Sizes and capacities. The screw machine size and capacity shall meet the requirements of Table I. Options shall be provided only when specified by the procuring activity (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

TABLE I

SIZES AND CAPACITIES

MACHINE CHARACTERISTIC	REQUIREMENT (Dimensions in Inches)						
	Class 1		Class 2				
	1/2 inch	3/4 inch	3/4 inch	1 1/4 inch	1 5/8 inch	2 inch	2 3/8 inch
Collet capacity - maximum diameter round bar, inches	1/2	3/4	3/4	1 1/4	1 5/8	2	2 3/8
Turning length - maximum, inches	1 1/2	1 1/2	3 1/2	3 1/2	3 1/2	3 1/2	3 1/2
Spindle speeds:							
Total range, RPM - 2 speed drive	34 to 8665	34 to 6060	20 to 5018	14 to 3500	14 to 2900	14 to 2484	14 to 2484
Number of speeds - High speed	18	17	18	18	17	17	17
Second speed	15	15	15	15	15	15	15
Spindle motor horsepower	3	3	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2
Optional spindle speeds:							
Total range, RPM - 4 speed drive	6 to 8665	6 to 6060	4 to 5018	3 to 3500	3 to 2900	3 to 2484	3 to 2484
Number of speeds - High speed	18	17	18	18	17	17	17
Second speed	11	11	10	10	10	10	10
Third speed	15	15	15	15	15	15	15
Fourth speed	165	165	150	150	150	150	150
Spindle motor horsepower with optional speeds	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2	7 1/2
Drive shaft speeds, RPM	120/240	120/240	120	120	120	120	120
Drive sleeve speeds, RPM	120/240	120/240	120/240	120/240	120/240	120/240	120/240
Bar feed in inches, single movement,							
drive sleeve at 120 RPM	2	2	4	4	4	4	4
drive sleeve at 240 RPM	1	1	2 3/4	2 1/2	2 1/2	2 1/4	2 1/4
Cycle time in seconds:							
Drive shaft at 120 RPM	1 1/2 to 100	1 1/2 to 100	1 2/3 to 1000	1 2/3 to 1000	1 2/3 to 1000	1 2/3 to 1000	1 2/3 to 1000
Drive shaft at 240 RPM	3/4 to 50	3/4 to 50	-	-	-	-	-
Tool turret - number of index positions	6	6	6	6	6	6	6
Diameter of turret tool holes, inches	5/8	5/8	1	1	1	1 1/4	1 1/4
Longitudinal adjustment of turret slide, inches	1/2	1/2	1	1	1	1	1

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

3.6 Alignment tolerance. The screw machine shall conform to the alignment to tolerances of Table II when checked in accordance with 4.5.2.

TABLE II. Alignment Tolerances

	CLASS 1	CLASS 2
Turret slide ways parallel to spindle		
Horizontal check, spindle end forward	0 to 0.0005" in 12"	0 to 0005" in 18"
Vertical check, spindle end high	0 to 0.0015" in 12"	0 to 0.0015" in 18"
Alignment of cross slide ways with spindle. Sweep front and back ways	0.006" in 18" sweep	0.008" in 24" sweep
Turret tool holes aligned with spindle		
Horizontal check, toward rear toward front	0.0007" 0.0007"	0.0007" 0.0007"
Vertical check, high low	0.0007" 0.0007"	0.0007" 0.0007"

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

* 3.7 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical system shall conform to the applicable requirements of NFPA - No. 79, except that a single disconnect means shall disconnect all ungrounded conductors. An identified terminal suitable for connecting the size grounding conductor specified in Table 204A of NFPA-79 shall be provided. The main drive motor shall be of the ball or roller bearing type, continuous duty, of the horsepower rating of Table I, and shall conform to the requirements of a dripproof enclosure. The machine shall operate on 230/460 volt, 3 phase, 60 Hz power, and unless otherwise specified (see 6.2.1), shall be initially wired for operation on 230 volts.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine.

3.8.1 Special tools. Tools designed by the machine tool builder and normally furnished to facilitate operation and maintenance of the machine shall be furnished.

3.8.2 Coolant system. The coolant system shall include a pump, reservoir, all necessary piping and hose to deliver coolant to the cutting tools, with means for directing and regulating the coolant flow. The reservoir shall be constructed to facilitate draining and cleaning.

3.8.3 Splash guard. The splash guard shall enclose the work area of the machine to confine chips and coolant, shall have a transparent section for viewing the machining operation, and shall be readily removeable for access to the work area.

3.8.4 Work light. A shielded lamp shall be provided on the machine to illuminate the work area.

3.8.5 Stock stands and tube. The stock stands shall be designed to support the stock tube and additional bars. Provisions shall be made in the stand bases for securing to the foundation. The stock tube shall be designed to suppress noise produced by the rotating bar and shall fully enclose a 12 foot long bar when it is inserted into the machine.

3.8.6 Change gears. A complete set of change gears shall be furnished to provide all spindle speeds and cycle speeds within the range of the machine.

3.8.7 Collets and feed fingers. A complete set of spindle collets and feed tube fingers shall be furnished as required (see 6.2.1) for gripping

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

round, square, and hexagon bars of diameters within the capacity of the machine. Unless otherwise specified under 6.2.1, round collets shall be in 1/32" increments, square and hexagon collets shall be in 1/16" increments.

3.9 Optional equipment. When specified (see 6.2.1), the following equipment shall be furnished with the machine.

3.9.1 Eight-hole tool turret. The eight hole, indexing tool turret shall be provided in addition to, or in lieu of, a six-hole turret, as specified in ordering data, for mounting round shank tools not less than 1 inch diameter.

3.9.2 Cross drilling attachment. The cross drilling attachment shall be a complete unit for mounting on the rear cross slide and shall consist essentially of a ball bearing mounted drill chuck, an electric drive motor and spindle brake.

3.9.3 Screw slotting attachment. The screw slotting attachment shall be a complete unit consisting essentially of an electric motor for driving a circular saw, a pick-up and feed mechanism, and cams.

3.9.4 Turret drilling attachment. The drill attachment shall be a complete unit consisting essentially of the drilling spindle for mounting in the tool turret, an electric drive motor, and drive mechanism. Maximum drill capacity shall be at least 3/32 inch diameter for Class 1 machines and 3/16 inch diameter for Class 2 machines.

3.9.5 Cam shaft accelerator for Class 2 machines. The cam shaft accelerator shall automatically accelerate the cam shaft to reduce noncutting time when indexing tools. The accelerator mechanism shall be electrically controlled, actuated by adjustable trips.

3.9.6 Longitudinal turning slide. The turning slide shall be a complete unit for mounting on the front cross slide. It shall be for turning straight and taper diameters within the bar capacity of the machine. Turning length shall be at least:

1-3/8 inches, Class 1 machines
3 inches, Class 2 machines

Tool holder shall take a square tool at least:

5/16 inch, Class 1 machines
1/2 inch, Class 2 machines

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

3.9.7 Single point threading attachment for Class 2 machines. The threading attachment shall be a complete unit consisting essentially of a tool holder mounted on the front cross slide having longitudinal movement, and shall be synchronized with and driven from the machine spindle. The attachment shall thread diameters up to the bar capacity of the machine, and at least two inches in length. The attachment shall be adjustable to at least 7^{0} in both directions for taper threading.

3.9.8 Turret tool coolant system for Class 2 machines. The coolant system shall be a complete unit for delivering cutting coolant through any tool mounted in the turret, when indexed to the cutting position.

* 3.9.9 Double grip collets, class 2 machines. Double grip collet sets, collets having two gripping surfaces contacting the stock the length of the collet grip surfaces shall be provided. Each collet shall be designed to hold stock sizes in fractional inch, decimal and metric sizes and cover the stock size capacity of the machine.

3.9.10 Elapsed time meter. The meter shall be a digital read-out, non-reset type, designed to resist the effects of vibration, voltage fluctuation, temperature and humidity. The meter shall be mounted on the machine and connected to record accumulated operating time of the machine to a total of at least 9999.9 hours.

3.10 Additional performance requirements. Performance features, tooling and equipment required in addition to, or in lieu of, the requirements herein shall be as specified (see 6.2.1).

3.11 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.12 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.13 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

Point of application
Service interval
Type of lubricant
Viscosity

3.14 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate or plates shall be securely attached to each machine. The nameplate(s) shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture
U S

3.15 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.16 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

specified (see 6.2.1), first article inspection shall comprise the examination in 4.4 and all test in 4.5. Failure of the item to pass any examination or test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5 through 4.5.2, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.7 through 3.16.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated in accordance with the machine tool builder's standard operating test procedure for warm-up and run-off checks. Run-off checks shall be made of all automatic functions of the machine by cycling in the automatic mode of operation. The cycle shall include operating the spindle speed change mechanism, indexing mechanism, built in attachments. Response to manually operated electrical controls shall be checked. The machine shall be operated automatically for not less than 30 minutes without a malfunction.

4.5.2 Accuracy test. The machine shall be checked in accordance with the machine tool builder's standard test methods to determine compliance with machine alignment tolerances in 3.6. Test instruments of the machine tool builder's choice may be used which shall have an accuracy traceable to the Bureau of Standards.

4.5.3 Electromagnetic interference control tests. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.11.

4.6 Inspection of packaging. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with the applicable requirements of MIL-M-18058. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

required level of preservation and packing, and any special marking requirements shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The automatic screw machines covered by this specification are intended for use in any machine shop for producing a variety of products, from bar stock. Machining operations include automatically turning, drilling, forming, threading, and slotting. The machines covered by this specification are of one type. The basic difference between classes is that Class 2 machines have more work length capacity, more horsepower, and more possible attachments than are available for Class 1 machines.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Class and size machine required (see 1.2).
c. First article approval, if required (see 3.1).
d. Specify additional safety and health requirements (see 3.2.1 and 6.4).
e. Painting, if different, and Machine Tool Light Gray, if required (see 3.3.4).
f. Optional spindle speeds, if required (see Table I).
g. Electrical equipment, if different, type motor enclosure, if different, voltage requirements, if different, and voltage for initial wiring, if different (see 3.7).
h. Standard equipment, if different (see 3.8).
i. Collet type and size increments, as required (see 3.8.7).
j. Optional equipment, as required (see 3.9).
k. Additional performance requirements, if required (see 3.10).
l. Electromagnetic interference control, if required (see 3.11).
m. Type fungus proofing, if required (see 3.12).
n. Lubrication chart or plate requirement, if different (see 3.13).
o. Nameplate, if different (see 3.14).
p. First article inspection, if different (see 4.2).
q. Quality conformance inspection, if different (see 4.3).
r. Electromagnetic interference control test requirements, if different (see 4.5.3 and 6.4).
s. Required level of preservation, packing and marking (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required under 3.1, it should be tested and approved under the appropriate provisions of 7-104, 55 of the Armed Services Procurement Regulation (ASPR). The first article should be a production model or it may be standard production item from the contractors current inventory. The Contracting Officer should include specific instructions in all procurement instruments, regarding arrangement for examinations, test and approval of the first article.

6.4 Safety and health determination. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

6.5 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contactors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH
DLA - GS

User activities:
Navy - AS, MC

Preparing activity:
DLA - IP

Project Number:
3416-0164

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-23378C

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.

FOLD

HEADQUARTERS
DEFENSE INDUSTRIAL PLANT EQUIPMENT CENTER
MEMPHIS, TENNESSEE 38114

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE, $300

POSTAGE AND FEES PAID
DEFENSE LOGISTICS AGENCY
DoD-304



Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-TF
Memphis, Tennessee 38114

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:35Z
Check the source to verify that this is the current version before use.